Report Sections: Human Validation Study

Human Validation Study Design

The human study checks whether the automated Composite Usability Score (CUS) ranks the three chatbots the way real users do. Each participant tests both chatbots once and rates each with the System Usability Scale (SUS).

Chatbots under test. Chatbot A (Llama 3.1 8B) and Chatbot B (Mistral 7B), both run locally through Ollama with temperature 0.7, as in the automated runs. Rasa is scored automatically but is not part of the human study, because it could not be hosted for testers.

Task script. Every participant sends the same six messages to each chatbot: a vegetarian diet plan, a three-course Italian menu, a $300 smart-home starter kit, a resume-bullet improvement, an explanation of compound interest, and a misspelled order query ("wehre is my odrer nubmer 4521"). The first three build on one another, so the script exercises context retention. The last one exercises error handling. A web walkthrough page gives the messages, tracks progress and unlocks the survey only after all six are sent.

Instrument. After each chatbot, the participant completes the 10-item SUS (Brooke, 1996) on a 1 to 5 scale in a Google Form. Odd items score as (answer - 1), even items as (5 - answer). The sum is multiplied by 2.5, giving 0 to 100. A score of 68 is commonly cited as the average benchmark (Sauro), and we report it as a reference point, not a pass mark.

Order effects. The starting chatbot alternates across participants (A then B, or B then A) so that no bot is always tested first.

Data quality. Responses with straight-lined answers or a missing item are flagged in the scoring sheet and reviewed before analysis.

Analysis plan.

1. Per bot: n, mean SUS, SD, min, max and the share of participants scoring 68 or above.
2. Compare the SUS ranking of the two bots with their CUS ranking, and report the SUS difference with a Welch t-test as supporting detail.
3. Report the rank agreement as a descriptive result.

Limitations. With two chatbots in the human study, any correlation between CUS and SUS rests on only two points. A Pearson or Spearman coefficient from two points cannot support any claim, so we present the rank comparison as the main result and state the coefficient only as supporting detail. The target of 30 participants per bot supports a stable per-bot SUS mean and a reasonable comparison of the two bots. It does not turn two CUS values into a correlation. Because Rasa is excluded, the human study also cannot check the low end of the CUS scale. The sample is also a convenience sample of students, family and friends, so results describe this group and should not be generalised to all users. CUS values are provisional until the automated re-scoring is finalised.

Recruitment Plan

We aim for 30 participants, each rating both chatbots, which gives 30 SUS scores per bot. Each session takes about 10 to 15 minutes.

Group	Target	Channel	Notes
Classmates	15 to 20	Class WhatsApp group	Main source; technically literate
Family and friends	4 to 6	Direct message	Less technical; widens the user mix
Project teammates	Remaining	Direct message	Counted separately, see below

Teammate bias. Members A and B built or configured the automated pipeline and know the expected behaviour of each bot. Their ratings are collected but reported separately, and the main analysis is run both with and without them.

Participation. Participation is voluntary and anonymous. The form collects no name or email, and participants may stop at any time. We tell them the study is for a university project and that replies can take several seconds, so they rate what actually happened.

Order rotation. Each participant is assigned a starting chatbot in turn (A, B, A, B and so on), recorded when the invitation is sent.

Data handling. Responses stay in a private Google Sheet shared only within the team. Raw rows are not published. The report gives only aggregated results.

Timeline. The chatbots run in a Colab session that must stay open while people test, so testing happens in announced evening windows of two to three hours. Each window uses fresh links. After the last window the sheet is closed and analysed.

Results

Sample. The form closed with 61 responses: 31 for Chatbot A (Llama 3.1 8B) and 30 for Chatbot B (Mistral 7B). SUS was scored with Brooke's method (odd items: answer − 1; even items: 5 − answer; sum × 2.5, range 0–100).

	Llama 3.1 8B	Mistral 7B
Responses	31	30
Mean SUS	71.0	70.9
Standard deviation	14.7	21.3
Median	72.5	75.0
Range	47.5 – 100	32.5 – 100
Scores at or above 68	20 of 31	19 of 30

Main finding. The two chatbots are statistically indistinguishable on perceived usability. The mean difference is 0.1 SUS points (Welch t-test p = 0.98; Mann-Whitney p = 0.89). Both sit just above the commonly cited benchmark of 68 (Sauro), which means average to slightly better than average usability. Mistral's scores are more spread out (SD 21.3 against 14.7): it has more testers who rated it very low (5 scores below 50, against 1 for Llama) as well as more who rated it very high.

Comparison with the automated score. The Composite Usability Score (CUS) ranks Llama (59.64) above Mistral (51.54), an 8-point gap. The SUS order is nominally the same (Llama 71.0, Mistral 70.9), but the gap is only 0.1 points and not significant, so testers effectively rated the two bots the same and the 8-point CUS gap is not confirmed. With two chatbots we cannot compute a meaningful correlation, so we report the rank comparison only. The honest reading is that the automated metrics and human perception agree that both bots are usable, but the CUS gap is not confirmed by testers. A likely reason is that the CUS includes technical measures such as response time and hedging, while SUS reflects overall impression after a short conversation.

What drives the CUS gap. The CUS is the equal-weighted mean of five normalized metrics (RT = 100 − response time in ms / 100, floored at 0; ER = 100 − error rate). Mean over three runs:

Metric (0–100)	Llama	Mistral	Difference
Context retention	33.3	11.1	+22.2

Metric (0–100)	Llama	Mistral	Difference
Response time	0.0	6.2	−6.2

Helpfulness	100.0	84.4	+15.6
Lexical diversity (MATTR)	73.8	76.0	−2.2
Error rate (inverted)	91.1	80.0	+11.1
CUS	**59.64**	**51.54**	**+8.1**

Llama's lead comes from context retention, helpfulness and error rate. Response time adds almost nothing for either bot: both take about 8 to 18 seconds, so the metric sits at its floor and Llama's slowness does not cost it in the CUS. Testers did notice the slowness, but they still rated usability the same.

SUS in more detail. The 95% confidence interval for the SUS difference (Llama minus Mistral) is −9.3 to +9.6 points, so a difference of up to about 9 points in either direction cannot be ruled out; the effect size is negligible (Cohen's d about 0.01). No single SUS item differs significantly between the bots (all Mann-Whitney $p > 0.18$). The largest gaps are "not cumbersome" (Llama 3.06 vs Mistral 2.53 out of 4) and "would use frequently" (Mistral 2.80 vs Llama 2.55). Learnability (72.2 vs 71.7) and usability (70.8 vs 70.7) subscale scores are the same. The spread does differ: Mistral's scores vary significantly more (Levene $p = 0.04$), so it divides testers more than Llama does.

Why the CUS and SUS differ. The two scores measure different things, so a gap between them is expected and is not an error in either. We see four reasons.

1. **Different constructs.** The CUS scores model behaviour on a fixed 87-turn script (context retention, speed, helpfulness, diversity, errors). SUS scores the overall experience of a short chat, and both chatbots were shown in the same chat interface. Testers sent six messages and did not check a vegetarian constraint over 30 turns.
2. **Coarse context retention.** It uses 6 checkpoints per run, all testing the same vegetarian constraint. One checkpoint is worth 3.3 CUS points, and context retention accounts for about half of the 8-point gap. The failures are genuine (for example carbonara, prosciutto and ham suggested to a vegetarian), but the metric is narrow.
3. **Response time sits at its floor.** It is normalized as 100 − ms / 100, so any reply slower than 10 seconds scores 0. Both LLMs take 8 to 18 seconds on the Colab GPU. The table below shows how the CUS would change under gentler scoring.
4. **An unvalidated judge.** The helpfulness judge has not been checked against human ratings (kappa = 0.0 on 3 turns), so the helpfulness gap is provisional.

Response-time scoring	Llama CUS	Mistral CUS	Gap
Current (0 at 10 s)	59.6	51.5	8.1
Linear, 0 at 20 s	62.1	59.9	2.1
Linear, 0 at 30 s	67.9	63.4	4.5
Response time dropped	74.6	62.9	11.7

Llama stays ahead in every variant, but the size of the gap depends strongly on how response time is scored, which is a limitation of the CUS design. The SD values of 0.66 and 1.57 only show that repeating the same script gives the same scores; they do not show that the gap would hold on other scripts or topics.

Tester comments. Of 17 Llama comments, 5 said the replies were slow, which matches its longer response time in the automated RT metric. Of 12 Mistral comments, 3 praised its speed or time saving and 1 said it answered late. Two Mistral comments praised consistency and context handling, although its automated context-retention score is the lowest (11.1%); a short six-message chat does not stress context the way the 30-turn script does. Some testers found Llama's answers more detailed, and some preferred Mistral's shorter answers.

Limitations. (1) Only two chatbots, so no correlation can be calculated; Rasa was excluded because it ran locally and could not be offered to testers. (2) The form had no tester ID, so scores cannot be paired by person; the two groups are treated as independent. (3) Testers used a shared Colab-hosted setup, and slow replies on the free GPU may have lowered scores for both bots. (4) Team members and friends make up part of the sample (teammate bias); results will be re-reported with and without them once timestamps are matched. (5) One early test response dated 9 Oct is excluded from the 61. (6) A sample of about 30 per bot gives wide confidence intervals, so a difference smaller than roughly 13 SUS points (at 80% power) could not have been detected.